MILITARY STANDARD

RAILCAR LOADING OF HAZARDOUS MATERIALS

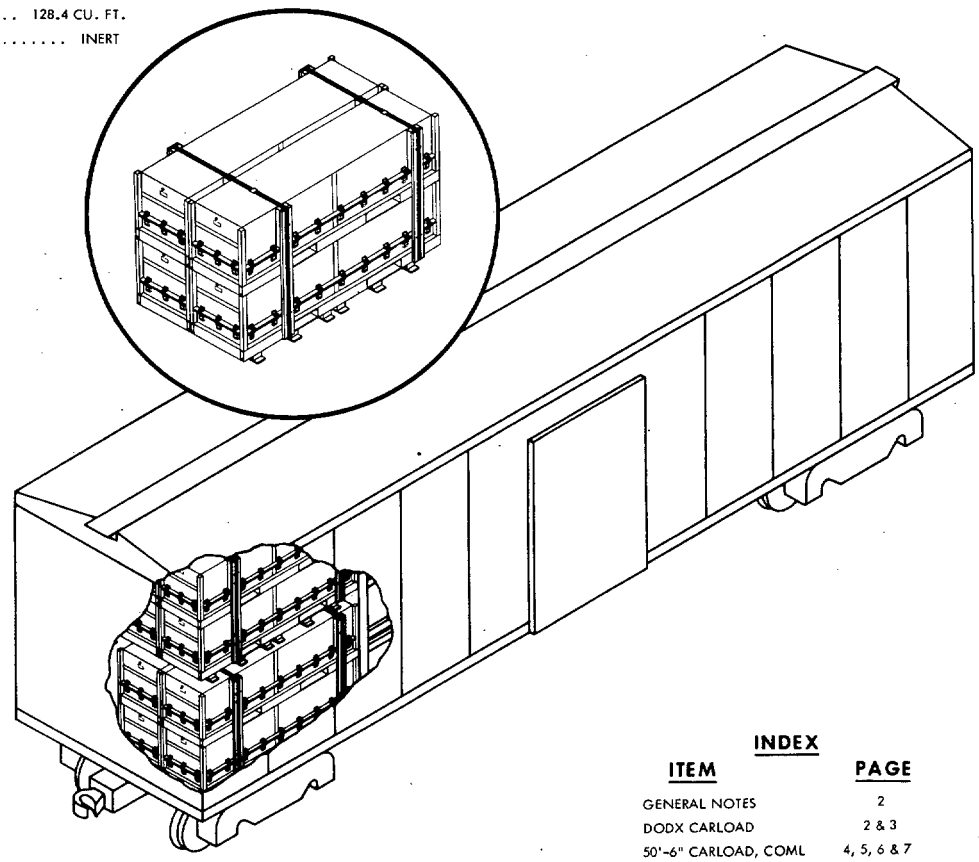
WING ASSEMBLY MK 7 MOD 2
AND FIN ASSEMBLY MK 32 MOD 2
FOR GUIDED WEAPON MK 5 MOD 4 (WALLEYE II)
IN CONTAINER CNU-150/E
FLEET ISSUE UNIT LOAD

MIL-STD-1325-172
(NAVY)

3 JANUARY 1975

UNIT LOAD DATA

UNIT LOAD DRAWING WR-54/255
WEIGHT (APPROX) ..... 1627 LBS
DIMENSIONS .. 85 3/16 L x 46 3/4 W x 55 11/16 H (STACKING HEIGHT 55 3/16)
CUBE ..... 128.4 CU. FT.
DOT HAZARD CLASSIFICATION ..... INERT



INDEX

ITEM	PAGE
GENERAL NOTES	2
DODX CARLOAD	2 & 3
50'-6" CARLOAD, COML	4, 5, 6 & 7

NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, TRUCKLOADING AND CONTAINERLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS MIL-HDBK-236 (NAVY)

FSC 8140

AUTHORIZED AND RELEASED FOR GENERAL USE.

W S Gray NWHL 10/10/74
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

J C Kelly 10/10/74
SIGNATURE AIRSYSCOM BY DIRECTION DATE

APPROVED BY BUREAU OF EXPLOSIVES
NOT APPLICABLE

SIGNATURE SUPERVISOR, MILTIARY & INTERMODEL SERVICES

ORIGINATOR Charles McPhee 10/10/74
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
N A D EARLE, NEW JERSEY

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GENERAL NOTES

1. FOR GENERAL INFORMATION CONCERNING ORDERING, INSPECTING, AND PREPARING CARS, AND FOR DUNNAGING MATERIALS, DESIGN, AND INSTALLATION OF DUNNAGE SEE THE GENERAL DOCUMENT MIL-STD-1325 (NAVY) "RAILCAR LOADING OF HAZARDOUS MATERIALS."
2. WHEN PLANNING SHIPMENTS ORDER THE MINIMUM NUMBER OF CARS OF THE CAPACITY REQUIRED FOR THE SHIPMENT. UTILITY LOADER CARS SHALL BE SERIES DODX 28000.
3. LOADING PLANS SHOWN ARE FOR DODX UTILITY LOADER CAR WITH 50 FT 6 INCHES INSIDE LENGTH, 107 3/4 INCHES INSIDE WIDTH BETWEEN RAILS (111 INCHES INSIDE WIDTH BETWEEN SIDE WALLS), AND COMMERCIAL BOXCARS WITH 50 FT 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH.
4. THE "LOAD LIMIT" OF A CAR MUST NOT BE EXCEEDED NOR SHOULD THE RAILCAR BE LOADED SO THAT MORE THAN ONE-HALF OF THE "LOAD LIMIT" IS CARRIED BY ONE SET OF TRUCKS.
5. IF END WALLS OF CARS ARE NOT SQUARE THEY MUST BE SQUARED OFF BEFORE STARTING TO LOAD CAR.
6. THE LOAD CONSISTS OF WING ASSEMBLIES MK 7 MOD 2 AND FIN ASSEMBLIES MK 32 MOD 2 FOR GUIDED WEAPON MK 5 MOD 4 (WALLEYE II) IN CONTAINER CNU-150/E PALLETIZED IN ACCORDANCE WITH WR-54/255.
7. THE UNIT LOADS ARE HANDLED AND LOADED WITH A SUITABLE FORK LIFT TRUCK. TO FACILITATE HANDLING THE UNIT LOAD FROM THE END, FORK TINE EXTENSIONS MK 8 MOD 0 AS DEPICTED IN NAVORD OP 2173 VOLUME 1 (THIRD REVISION) MAY BE USED.
8. UNLESS OTHERWISE SPECIFIED NAILING SHALL BE IN ACCORDANCE WITH MIL-STD-1325 (NAVY).
9. ALL STRAP JOINTS SHALL BE CRIMP-TYPE JOINTS MADE WITH A CRIMPING TOOL THAT DOES NOT CUT THE SEAL OR STRAP, RATHER THAN A TOOL THAT MAKES A NOTCH-TYPE JOINT. THE STRAPPING MUST BE IN ACCORDANCE WITH NAVSUP NOTICE 4642 DATED 21 APRIL 1971.
10. APPLICABLE MATERIAL SPECIFICATIONS:

    DUNNAGE LUMBER - FED. SPEC MM-L-751

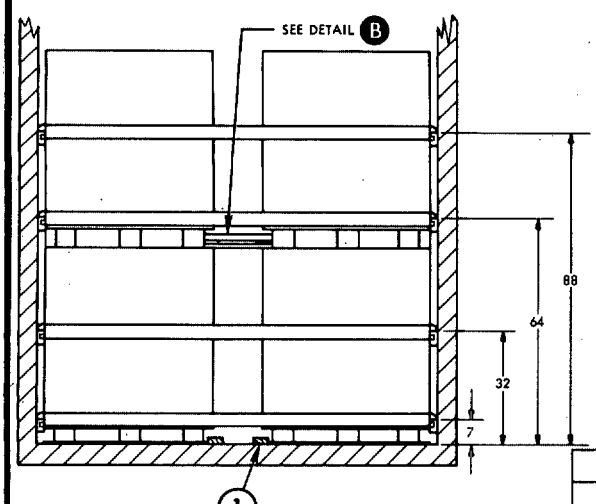
    NAILS - FED. SPEC FF-N-105

    STRAPPING - FED. SPEC QQ-S-781, TYPE I, HEAVY DUTY, CLASS A, DRY (UNLUBRICATED).

    SEALS - FED. SPEC QQ-S-781, STYLE III, HEAVY DUTY
11. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS INSIDE THE CAR IN AN ACCESSIBLE AREA, CLOSE AND SEAL BOXCAR DOORS.

50 FT. 6 IN. BOXCAR, DODX

1. THE CARLOAD CONSISTS OF 24 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. A DETAILED DESCRIPTION AND OPERATING INSTRUCTIONS FOR THE UTILITY LOADER ARE CONTAINED IN OP 1750.
3. TO PREVENT UNUSED "DF" EQUIPMENT FROM BECOMING DISLODGED DURING TRANSIT OF DODX CARS SECURE IT AT ANY LOCATION IN THE BOXCAR WHICH WILL NOT INTERFERE WITH UNLOADING.
4. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN DODX BOXCARS THE SAME PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE. ANY BAYS OR PORTION THEREOF MAY BE USED PROVIDING THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR ARE COMPLIED WITH, SEE MIL-STD-1325 (NAVY). EACH CROSSMEMBER WILL BE USED IN SUCH A MANNER THAT IT WILL RETAIN NOT MORE THAN 2000 LBS OF THE LADING.



SECTION A-A

SHOWING LOCATION OF WALL MEMBERS AND DOORWAY MEMBERS USED FOR CROSSMEMBERS

*W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR.

** PIECE 3 NAILS TO PIECE 4 WITH 10d NAILS, CLINCHED, 3 PER JOINT, AND TO PIECE 2 WITH 6d NAILS, 1 PER JOINT.

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
4	CROSS BRACE	2 x 6 x W*	36	SEE 3	-	-
3	SWAY BRACE LONGITUDINAL	1 x 4 x 36	24	2, 4	SEE NOTE **	
2	SWAY BRACE SUPPORT	1 x 4 x (W+8)*	24	SEE 3	-	-
1	SLEEPER	2 x 4 x 70	12	CAR FLOOR	2 PER FOOT	16d

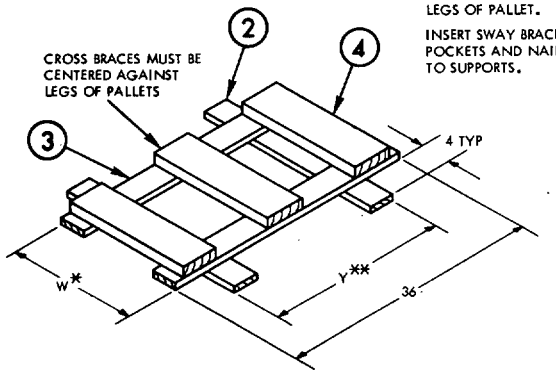
LIST OF MATERIALS AND NAILING DATA

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*W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR.

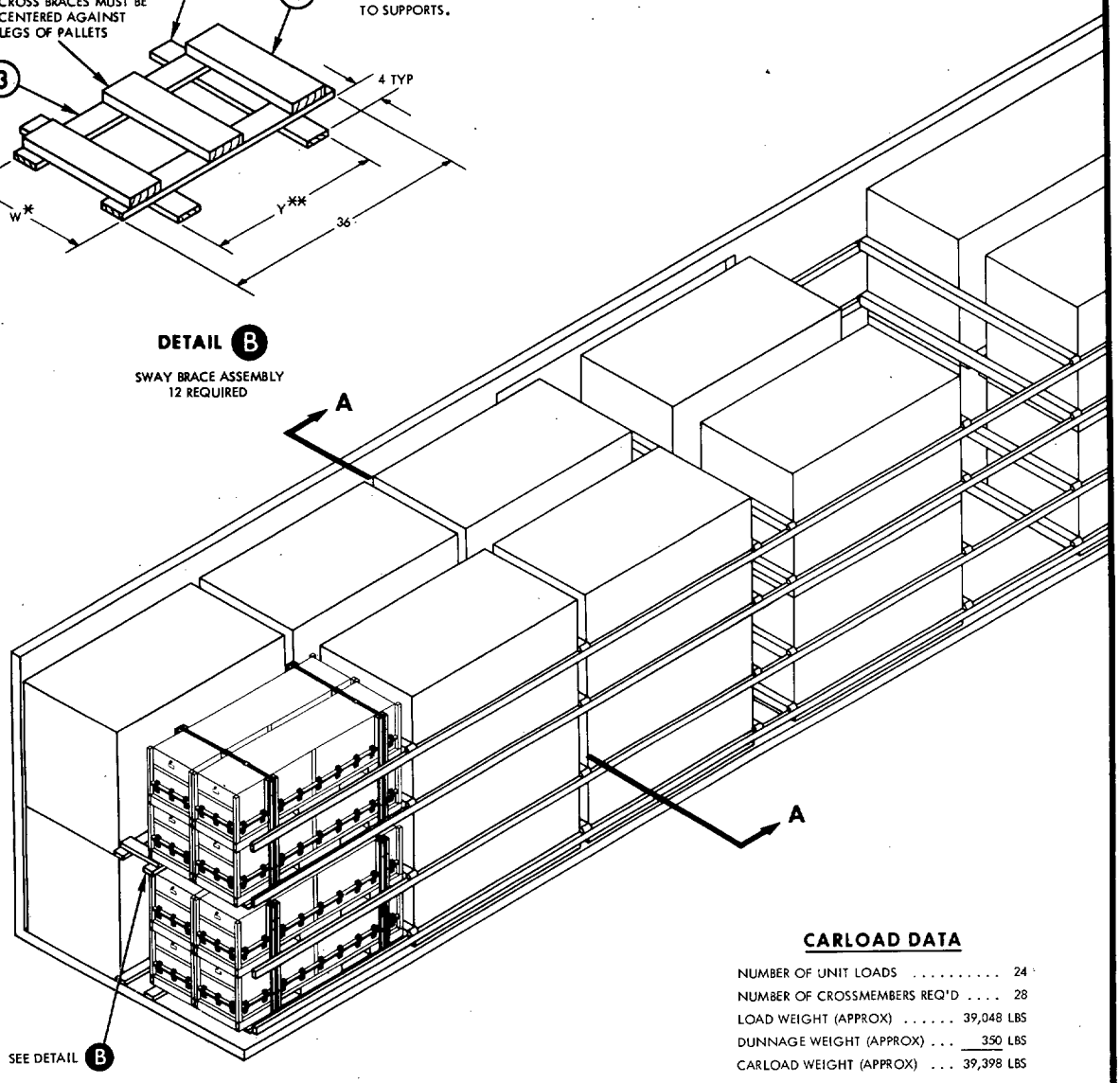
**Y = DISTANCE BETWEEN INSIDE OF OUTER LEGS OF PALLET.

INSERT SWAY BRACE SUPPORTS IN FORK POCKETS AND NAIL SWAY BRACE FRAME TO SUPPORTS.



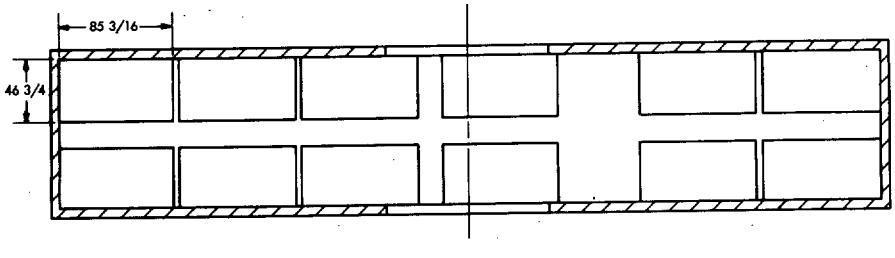
DETAIL B

SWAY BRACE ASSEMBLY
12 REQUIRED



CARLOAD DATA

NUMBER OF UNIT LOADS	24
NUMBER OF CROSSMEMBERS REQ'D	28
LOAD WEIGHT (APPROX)	39,048 LBS
DUNNAGE WEIGHT (APPROX)	350 LBS
CARLOAD WEIGHT (APPROX)	39,398 LBS

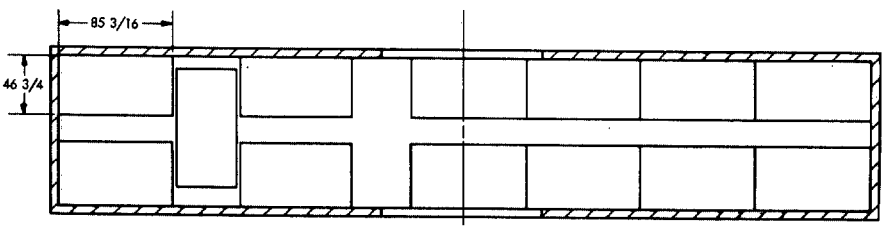


LOADING PLAN

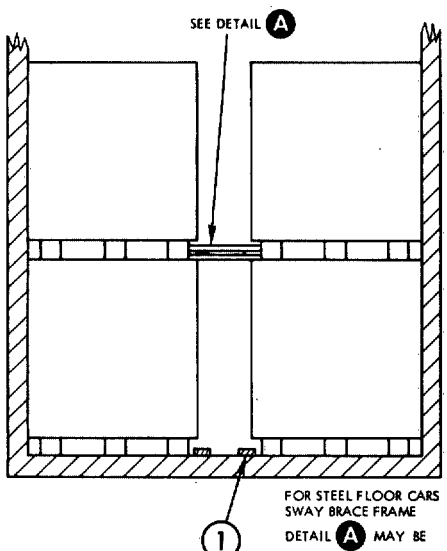
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50 FT 6 IN. BOXCAR, COMMERCIAL

1. THE CARLOAD CONSISTS OF 26 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING. THE CAR MAY ALSO BE LOADED IN ACCORDANCE WITH THE ALTERNATE LOADING PLAN ON PAGE 7 WHICH ELIMINATES THE CROSSWISE STACK ALONG WITH ITS SEPARATOR GATES AND SWAY BRACE ASSEMBLIES, BUT REQUIRES THE ADDITION OF HORIZONTAL AND VERTICAL TIE BARS NAILED TO THE CENTER GATE STRUTS (SEE DETAIL H).
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH WR-52/100. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE CENTER GATE HEIGHT SHOULD BE ADJUSTED AS REQUIRED. THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR MUST BE COMPLIED WITH, SEE MIL-STD-1325 (NAVY). THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 10 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 10 FT WIDE.
3. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE ALSO APPLICABLE TO BOXCARS EQUIPPED WITH PLUG TYPE DOORS. STACKS WITH MORE THAN HALF OF THE UNIT LOAD IN THE DOORWAY AREA MUST BE UNITIZED FOR DOORWAY PROTECTION WITH TWO LATERALLY APPLIED 1 1/4" STEEL STRAPS PER STACK, EACH TENSIONED AND SEALED WITH TWO DOUBLE CRIMPED SEALS AND A TOP SWAY BRACE CONSISTING OF A STRAPPING BOARD AND CROSS BRACE, TWO PER STACK, (SEE DETAIL F) MUST BE INSTALLED BETWEEN THE ROWS OF THE DOORWAY STACKS. STRAP IS SECURED TO STRAPPING BOARD WITH TWO 1 1/4" STAPLES. DIMENSIONAL LUMBER DOORWAY PROTECTION IS NOT REQUIRED WHEN PLUG DOOR EQUIPPED BOXCARS ARE USED, EXCEPT WHEN CAR HAS A COMBINATION OF PLUG DOOR AND CONVENTIONAL SLIDING DOOR. THEN DIMENSIONAL LUMBER DOORWAY PROTECTION IS REQUIRED FOR THE CONVENTIONAL DOOR. SECURELY CLOSE DOORS AND WIRE TOGETHER WITH A STRONG FLEXIBLE STEEL WIRE INSERTED THROUGH THE HOLES IN THE DOOR LATCH ASSEMBLY ONE OR MORE TIMES AND THE WIRE ENDS TWISTED TOGETHER.



LOADING PLAN



SECTION B-B

FOR STEEL FLOOR CARS SWAY BRACE FRAME DETAIL A MAY BE USED IN PLACE OF SLEEPERS PIECE 1.

*W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR.

**CLINCHED

***PIECE 3 NAILS TO PIECE 4 WITH 10d NAILS, CLINCHED, 3 PER JOINT, AND TO PIECE 2 WITH 6d NAILS, 1 PER JOINT.

PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NUMBER	SIZE
					NAILS	
18	CENTER GATE HOLDDOWN	2 x 4 x 60	2	17	7	16d
17	CENTER GATE HOLDDOWN	2 x 4 x 60	2	12	7	10d
16	STRUT	4 x 4 x WEDGE FIT	16	12	2 PER JOINT	16d
15	HORIZONTAL DOORWAY MEMBER	2 x 4 x DOOR WIDTH	8	14	2 PER JOINT	10d
14	VERTICAL DOORWAY MEMBER	2 x 3 x 96	4	DOOR POST	2 PER FOOT	20d
13	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH-1	8	12	3 PER JOINT	10d
12	CENTER GATE VERTICAL	2 x 6 x 96	8	SEE 11	-	-
11	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH-1	10	12	3 PER JOINT	10d
10	SWAY BRACE STRUT	2x4x CUT TO FIT	16	9	2 PER JOINT	16d TOENAIL
9	SWAY BRACE HORIZONTAL	2 x 4 x 46 1/2	16	8	3 PER JOINT	10d
8	SWAY BRACE VERTICAL	2 x 4 x 96	8	-	-	-
7	SEPARATOR GATE LOWER HORIZONTAL	2 x 4 x 73	2	SEE 5	-	-
6	SEPARATOR GATE UPPER HORIZONTAL	2 x 4 x CAR WIDTH-1	2	SEE 5	-	-
5	SEPARATOR GATE VERTICAL	1 x 4 x 115	8	6 & 7	3 PER JOINT	*** 10d
4	CROSS BRACE	2 x 6 x W*	36	SEE 3	-	-
3	SWAY BRACE LONGITUDINAL	1 x 4 x 36	24	2, 4	SEE NOTE **	
2	SWAY BRACE SUPPORT	1 x 4 x (W+8)*	24	SEE 3	-	-
1	SLEEPER	2 x 4 x 70	12	CAR FLOOR	2 PER FOOT	16d

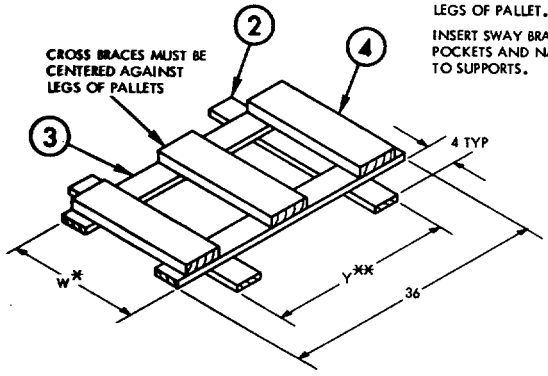
LIST OF MATERIALS AND NAILING DATA

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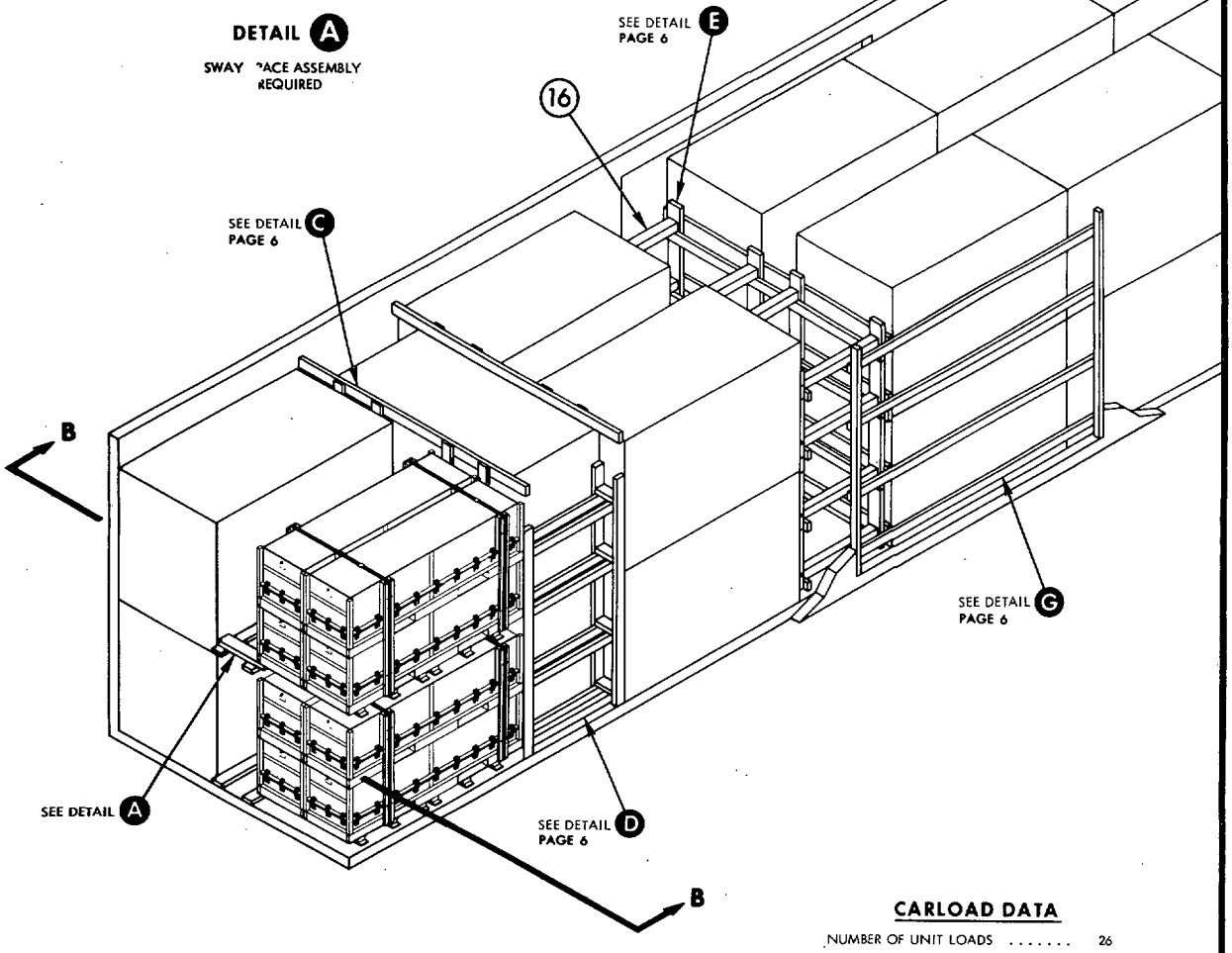
**Y = DISTANCE BETWEEN INSIDE OF OUTER LEGS OF PALLET.

INSERT SWAY BRACE SUPPORTS IN FORK POCKETS AND NAIL SWAY BRACE FRAME TO SUPPORTS.



DETAIL A

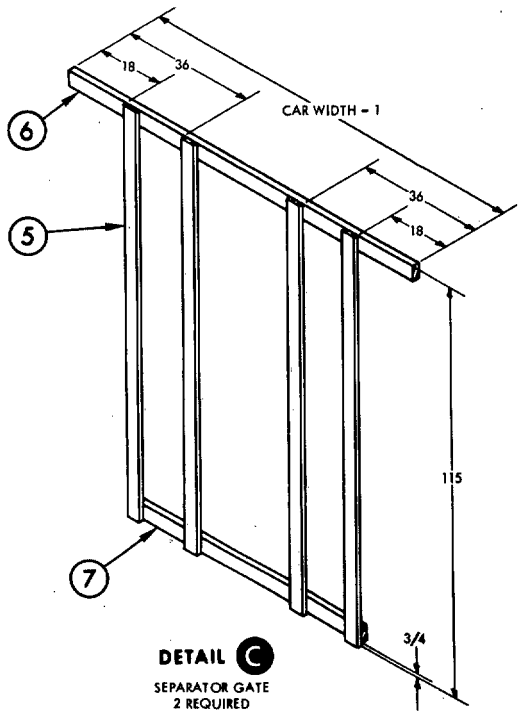
SWAY BRACE ASSEMBLY REQUIRED



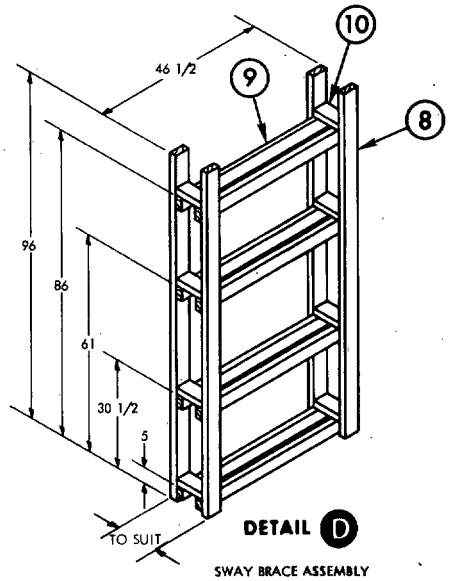
CARLOAD DATA

NUMBER OF UNIT LOADS	26
LOAD WEIGHT (APPROX)	42,302 LBS
DUNNAGE WEIGHT (APPROX)	1,649 LBS
CARLOAD WEIGHT (APPROX)	43,951 LBS

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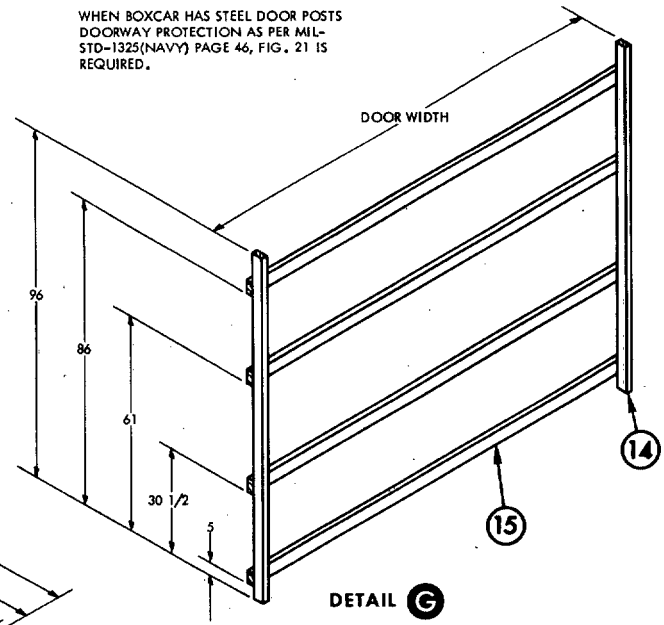


DETAIL C
SEPARATOR GATE
2 REQUIRED

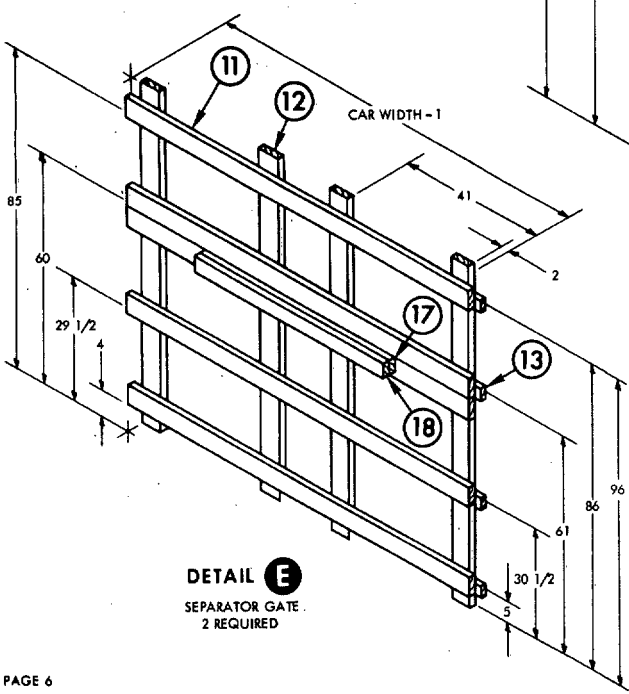


DETAIL D
SWAY BRACE ASSEMBLY
2 REQUIRED

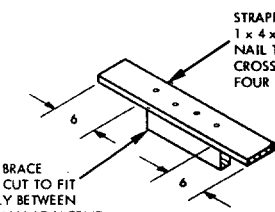
WHEN BOXCAR HAS STEEL DOOR POSTS DOORWAY PROTECTION AS PER MIL-STD-1325(NAVY) PAGE 46, FIG. 21 IS REQUIRED.



DETAIL G
DOORWAY PROTECTION
2 REQUIRED



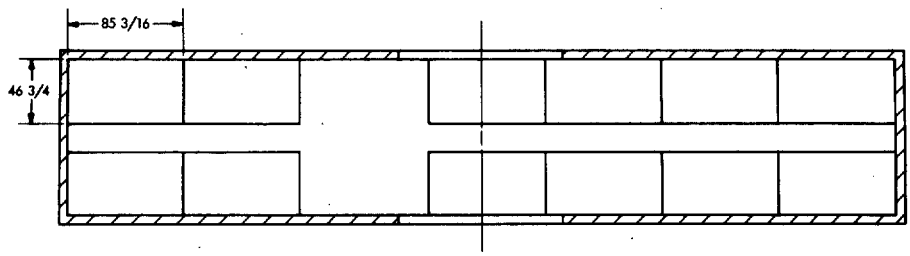
DETAIL E
SEPARATOR GATE
2 REQUIRED



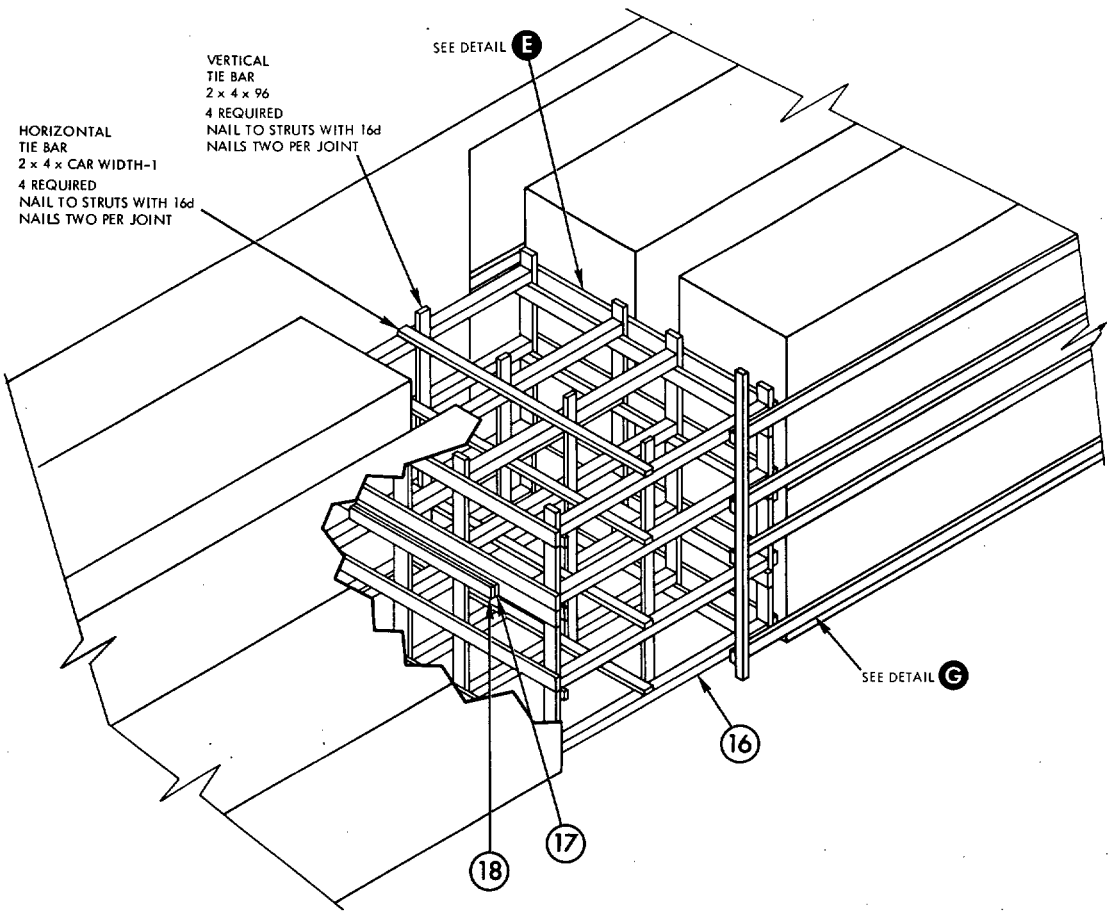
CROSS BRACE
2 x 4 x CUT TO FIT
SNUGLY BETWEEN
LATERALLY ADJACENT
PALLETS.

DETAIL F
TOP SWAY BRACE

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ALTERNATE LOADING PLAN



DETAIL H

ALTERNATE CARLOAD DATA

NUMBER OF UNIT LOADS	24
LOAD WEIGHT (APPROX)	39,048 LBS
DUNNAGE WEIGHT (APPROX)	1,866 LBS
CARLOAD WEIGHT (APPROX)	40,914 LBS

REVIEW ACTIVITY
NAVY - OS, AS

PREPARING ACTIVITY
NAVY - OS
(PROJECT NO. 8140-N250)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-STD-1325-172 (NAVY)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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